nccl-tests 1-node collective summary

- Generated: 2026-08-06 17:03:17
- Nodes: node5500, node5501, node5502 (each 8 x NVIDIA B200, single node, intra-node NVLink)
- Note: node5500-5502 are no longer in Slurm. The tables below are their historical baseline; the current hardware is the 7 new nodes covered in "New nodes" below.
- Config: 1 thread, 1 MiB-16 GiB, 5 warmup + 20 iters
- Collectives: sendrecv, reduce, broadcast, gather, scatter, reduce_scatter, all_gather, all_reduce, alltoall, hypercube
- Reference: MIT aicr-benchmarks `results_b200.md`, Table 1 (b0027, 8x B200, NVLink 5.0 / NVSwitch), busbw at 900 GB/s NVLink max

Converged busbw = busbw at the largest message size, best of out-of-place / in-place (matches the reference methodology). busbw (bus bandwidth) is the figure of merit.

Converged bus bandwidth by collective (GB/s)

Collective	node5500	node5501	node5502	Reference (b0027)	node5500 % of ref	node5501 % of ref	node5502 % of ref	Correctness
sendrecv	695.0	664.5	664.8	666	104%	100%	100%	PASS
reduce	673.7	685.8	671.7	701	96%	98%	96%	PASS
broadcast	707.7	677.6	702.4	691	102%	98%	102%	PASS
gather	755.7	717.8	700.7	717	105%	100%	98%	PASS
scatter	705.7	734.3	749.5	746	95%	98%	100%	PASS
reduce_scatter	667.2	691.3	715.3	695	96%	99%	103%	PASS
all_gather	721.3	673.2	667.0	684	105%	98%	98%	PASS
all_reduce	797.2	837.5	864.3	841	95%	100%	103%	PASS
alltoall	648.5	659.9	678.9	675	96%	98%	101%	PASS
hypercube	FAILED	FAILED	FAILED	—	—	—	—	FAIL

New nodes (node5600/5601/5602/5702/5800/5801/5802)

Run 2026-08-24 on the 7 new B200 nodes, same configuration as above (all 10 collectives, 8 GPUs, 1 MiB-16 GiB).

They match the nodes above — no separate per-collective tables needed. Comparing 7-node means against the node5500-5502 means, every collective agrees to within **2.8%** (largest gap: broadcast, -2.8%; next: reduce, +2.2%; all others within 1.3%). Absolute figures sit in the same band, e.g. all_reduce 832.9-852.6 GB/s (old mean 833.0), sendrecv 657.7-675.8 GB/s (old mean 674.8). Every node lands at **94-102% of the b0027 reference** on every passing collective.

Correctness: PASS everywhere except `hypercube`, which fails on all 7 nodes exactly as it did on all 3 old nodes — a known nccl-tests issue, not a node fault.

One mild outlier, not a fault. `node5602-c1` averages 1.4% below the 7-node mean and is the slowest node on the four root-anchored / ring collectives (broadcast, gather, scatter, reduce_scatter), worst case **-4.8%** on reduce_scatter. It still returns 94-96% of the reference — inside the spread the old nodes themselves showed (node5500 was at 95% on all_reduce and 96% on reduce). No action beyond a re-check if the same pattern shows up in a later run.

Bus bandwidth vs message size (out-of-place busbw, GB/s)

sendrecv

Message size	node5500	node5501	node5502
1 MiB	26.7	26.4	29.0
4 MiB	61.6	62.6	62.2
16 MiB	76.2	77.7	78.1
64 MiB	88.5	84.5	84.8
256 MiB	347.2	332.4	332.1
1 GiB	672.8	644.0	644.2
4 GiB	690.8	660.4	660.1
16 GiB	695.0	664.5	664.8

reduce

Message size	node5500	node5501	node5502
1 MiB	26.6	23.1	23.7
4 MiB	99.6	85.3	88.2
16 MiB	326.1	298.9	307.4
64 MiB	524.4	506.8	521.1
256 MiB	645.8	619.6	634.4
1 GiB	689.5	663.7	682.6
4 GiB	704.5	674.5	662.9
16 GiB	672.5	683.9	671.4

broadcast

Message size	node5500	node5501	node5502
1 MiB	24.7	26.0	26.1
4 MiB	84.3	94.5	97.3
16 MiB	294.7	305.4	313.9
64 MiB	517.0	496.5	514.7
256 MiB	638.2	611.1	630.0
1 GiB	675.5	643.9	664.3
4 GiB	690.4	659.9	684.3
16 GiB	707.7	677.6	701.8

gather

Message size	node5500	node5501	node5502
1 MiB	23.3	24.7	23.3
4 MiB	91.2	95.5	87.2
16 MiB	340.9	395.7	336.8
64 MiB	646.5	612.6	597.8
256 MiB	733.1	690.2	670.4
1 GiB	737.0	702.3	681.1
4 GiB	753.0	715.3	697.9

16 GiB	755.6	717.7	700.7

scatter

Message size	node5500	node5501	node5502
1 MiB	26.0	25.1	23.3
4 MiB	92.4	99.7	102.0
16 MiB	360.6	382.1	392.9
64 MiB	628.1	595.3	585.5
256 MiB	724.5	690.1	653.0
1 GiB	765.5	727.0	727.7
4 GiB	701.7	730.2	745.7
16 GiB	705.7	734.3	749.5

reduce_scatter

Message size	node5500	node5501	node5502
1 MiB	20.6	18.5	19.2
4 MiB	84.3	66.1	79.5
16 MiB	155.2	144.9	145.6
64 MiB	443.1	416.4	416.6
256 MiB	625.3	585.0	587.9
1 GiB	683.5	642.3	636.6
4 GiB	723.4	676.8	680.7
16 GiB	666.6	689.6	694.9

all_gather

Message size	node5500	node5501	node5502
1 MiB	16.8	15.7	16.9
4 MiB	71.2	65.1	71.4
16 MiB	136.2	137.1	137.1

64 MiB	407.5	416.0	410.6
256 MiB	568.3	579.8	573.8
1 GiB	603.5	616.3	607.2
4 GiB	639.7	649.1	642.2
16 GiB	656.2	665.6	659.0

all_reduce

Message size	node5500	node5501	node5502
1 MiB	35.4	34.1	37.2
4 MiB	119.6	125.3	126.6
16 MiB	252.8	266.1	268.8
64 MiB	399.0	422.2	420.0
256 MiB	622.0	659.1	658.0
1 GiB	697.1	730.6	733.6
4 GiB	787.0	826.2	853.1
16 GiB	797.2	836.2	863.5

alltoall

Message size	node5500	node5501	node5502
1 MiB	15.3	14.2	16.1
4 MiB	56.5	54.8	59.3
16 MiB	220.8	207.1	231.8
64 MiB	407.9	420.3	434.6
256 MiB	522.0	531.2	548.2
1 GiB	592.6	603.0	623.4
4 GiB	634.5	646.5	665.2
16 GiB	647.7	659.8	678.9

hypercube

No data (run failed or produced no rows).